

Google Cloud Cheat Sheet

Baseline: gcloud CLI 586, GKE 1.35 (Regular channel), Terraform hashicorp/google v8 (checked 2026-10-01). Examples use the Bookshelf scenario (my-project-id, us-central1, \$REGION). Quick reference only — verify against the official docs before production use. Irurueta Docs — Google Cloud section; see the full pages for details.

CHOOSING A COMPUTE SERVICE

Service	Best for	Scaling	Scale to zero	Notes
Cloud Run (services, jobs)	HTTP containers, APIs, batch	Concurrency (80), min/max instances	Yes (min-instances 0)	Source/image deploy; revisions + traffic split
Cloud Run functions	Event glue, single-purpose code	Same engine as Cloud Run	Yes	Renamed from Cloud Functions 2024-08-22; Eventarc triggers
GKE Autopilot	Kubernetes without node ops	HPA, per-pod billing	Pods yes, cluster no	Gateway API, Workload Identity Federation
GKE Standard	Custom nodes, GPUs, DaemonSets	Node pools, cluster autoscaler	Pool can hit 0	You manage node upgrades and sizing
Compute Engine	Lift-and-shift, OS control	Managed instance groups	No	OS Login, IAP tunnel, Spot; no konlet
App Engine	Existing apps only	Automatic	Standard: yes	Migrate to Cloud Run for new work

MANAGED DATABASES — ENGINE → GOOGLE CLOUD OPTION

Engine	Google Cloud option	Create (essentials)
PostgreSQL / MySQL / SQL Server	Cloud SQL (Enterprise, Enterprise Plus); AlloyDB for PostgreSQL (columnar engine, ScanNN, pgvector)	<code>gcloud sql instances create bookshelf-pg --database-version=POSTGRES_17 --edition=ENTERPRISE --tier=db-custom-2-8192 --region=\$REGION --availability-type=REGIONAL --database-flags=cloudsql.iam_authentication=on gcloud alloydb clusters create bookshelf --region=\$REGION ... + alloydb instances create primary --instance-type=PRIMARY --cpu-count=2</code>
Global SQL / NoSQL / wide-column	Spanner (editions, processing units); Firestore (Native); Bigtable	<code>gcloud spanner instances create bookshelf --config=regional-us-central1 --edition=ENTERPRISE --processing-units=100 --description=Bookshelf gcloud firestore databases create --database=bookshelf --location=nam5 gcloud bigtable instances create ...</code>
MongoDB	Firestore with MongoDB compatibility; Atlas; MongoDB on GKE	<code>gcloud firestore databases create --database=orders --location=nam5 --edition=enterprise</code>
Redis / Valkey / Memcached	Memorystore (Valkey, Redis Cluster, Redis); Memcached deprecated	<code>gcloud memorystore instances create cache-bookshelf --location=\$REGION --shard-count=1 --replica-count=1 --node-type=standard-small --endpoints=...</code>
Elasticsearch, Neo4j, Qdrant, Couchbase	Partner services (Elastic Cloud, Neo4j Aura, Qdrant Cloud, Capella) or Helm on GKE	Created in the partner console; connect over Private Service Connect or VPC peering

Auth & access: IAM database authentication (`--database-flags=cloudsql.iam_authentication=on, roles/cloudsql.instanceUser`); **Cloud SQL Auth Proxy** `cloud-sql-proxy my-project-id:us-central1:bookshelf-pg` or language connectors; private IP via Private Service Access / PSC; passwords belong in Secret Manager.

HIERARCHY & GCLOUD

Organization → folders → projects → resources; IAM and org policy inherit down. Project = ID (immutable), name, number. Scope: zone < region < multi-region; global. Deleted project: 30-day recovery.

```
gcloud init ; gcloud auth login
gcloud auth application-default login
gcloud config set project my-project-id
gcloud config configurations create dev
gcloud services enable run.googleapis.com
gcloud projects list --filter="labels.env=dev" --format="value(projectId)"
gcloud ... --impersonate-service-account=SA
```

No downloaded keys: use WIF or impersonation. Cloud Shell = zero install.

CLOUD RUN

```
gcloud run deploy catalog-api --source . \
--region $REGION --no-allow-unauthenticated \
--set-env-vars=LOG_LEVEL=info \
--set-secrets=DB_PASSWORD=db-pass:latest \
--min-instances=0 --max-instances=10 --concurrency=80
gcloud run deploy catalog-api --image IMG --no-traffic --tag green
gcloud run services update-traffic catalog-api --to-tags green=10
gcloud run jobs create nightly --image IMG
gcloud run jobs execute nightly --wait
gcloud run services add-iam-policy-binding catalog-api \
--member=serviceAccount:SA --role=roles/run.invoker
```

Billing: request-based vs. instance-based (`--no-cpu-throttling`). `--source = Cloud Build + Artifact Registry`. Cloud SQL: `--add-cloudsql-instances`.

FUNCTIONS & EVENTARC

```
gcloud run deploy fn --source . --function handler \
--base-image nodejs24
gcloud functions deploy fn --gen2 --runtime nodejs24 --trigger-http
gcloud eventarc triggers create t --location $REGION \
--event-filters="type=google.cloud.storage.object.v1.finalized" \
--event-filters="bucket=BKT" \
--destination-run-service=catalog-api --service-account=SA
```

Triggers: HTTP, Pub/Sub, Storage, Firestore, Audit Logs (CloudEvents, at-least-once).

ARTIFACT REGISTRY & CLOUD BUILD

```
gcloud artifacts repositories create bookshelf \
--repository-format=docker --location=$REGION
gcloud auth configure-docker $REGION-docker.pkg.dev
gcloud builds submit --tag \
$REGION-docker.pkg.dev/my-project-id/bookshelf/api:1.4.2
```

Image path: `REGION-docker.pkg.dev/PROJECT/REPO/IMAGE`. gcr.io is gone.

GKE

```
gcloud container clusters create-auto bookshelf \
--region $REGION --release-channel regular
gcloud container clusters get-credentials bookshelf \
--region $REGION # gke-gcloud-auth-plugin
gcloud container clusters upgrade bookshelf --master \
--cluster-version 1.35 --region $REGION
```

Channels: Rapid/Regular/Stable/Extended. **WIF principal:** `principal:/iam.googleapis.com/projects/NUM/locations/global/workloadIdentityPools/my-project-id.svc.id.goog/subject/ns/NS/sa/KSA`. **GatewayClasses:** `gke-l7-global-external-managed, gke-l7-regional-external-managed, gke-l7-ilb`.

CLOUD STORAGE

```
gcloud storage buckets create gs://BKT \
--location=$REGION --uniform-bucket-level-access \
--public-access-prevention
gcloud storage cp f gs://BKT/ ; gcloud storage rsync -r d gs://BKT/d
gcloud storage buckets update gs://BKT \
--lifecycle-file=lc.json --soft-delete-duration=7d
gcloud storage sign-url gs://BKT/o --duration=10m \
--impersonate-service-account=SA
```

Min. duration: Standard 0, Nearline 30 d, Coldline 90 d, Archive 365 d; **Autoclass** automates it. Lifecycle: `{"rule":[{"action":{"type":"SetStorageClass","storageClass":"NEARLINE"},"condition":{"age":30}}]}`. Static site: backend bucket behind an external ALB.

CLOUD CDN

```
gcloud compute backend-buckets create web \
--gcs-bucket-name=BKT --enable-cdn
gcloud compute backend-services update svc --global \
--enable-cdn --cache-mode=CACHE_ALL_STATIC
gcloud compute url-maps invalidate-cdn-cache-map --path "/"
```

Modes: `USE_ORIGIN_HEADERS, CACHE_ALL_STATIC, FORCE_CACHE_ALL`. Needs global external Application LB; Media CDN for video.

SECRET MANAGER & KMS

```
gcloud secrets create db-pass --replication-policy=automatic \
printf '%s' "$V" | gcloud secrets versions add db-pass --data-file=-
gcloud secrets versions access latest --secret=db-pass
gcloud secrets add-iam-policy-binding db-pass \
--member=serviceAccount:SA \
--role=roles/secretmanager.secretAccessor
gcloud kms keyrings create kr --location=$REGION
gcloud kms keys create k --keyring=kr --location=$REGION \
--purpose=encryption --rotation-period=90d \
--next-rotation-time=2027-01-01T00:00:00Z
```

Cloud Run `--set-secrets`; GKE Secret Manager add-on or External Secrets. Pin versions in prod.

PARAMETER MANAGER & CONFIGMAPS

```
gcloud parametermanager parameters create cfg \
--location=global --parameter-format=YAML
gcloud parametermanager parameters version s create v1 \
--parameter=cfg --location=global \
--payload-data-from-file=c.yaml
gcloud parametermanager parameters version s render v1 \
--parameter=cfg --location=global
kubectl create configmap cfg --from-file=c.yaml
```

Secret ref: `__REF__("/secretmanager.googleapis.com/projects/P/secrets/S/versions/1")`; role `parameterAccessor`. ConfigMap volumes refresh, env does not; Config Sync for GitOps.

VPC

```
gcloud compute networks create vpc1 --subnet-mode=custom
gcloud compute networks subnets create app --network=vpc1 \
--region=$REGION --range=10.0.0.0/24 \
--enable-private-ip-google-access
gcloud compute routers create r1 --network=vpc1 --region=$REGION
gcloud compute routers nats create nat1 --router=r1 \
--region=$REGION --auto-allocate-nat-external-ips \
--nat-all-subnet-ip-ranges
```

Global VPC, regional subnets; firewall policies (deny ingress by default). Private Google Access; PSA (peering) vs. PSC (endpoints); Shared VPC.

LOAD BALANCING, ARMOR & API GATEWAYS

HTTP(S) → **Application LB** (global/regional external, internal); TCP/SSL → **proxy Network LB**; UDP, client IP → **passthrough Network LB**.

```
gcloud compute security-policies create edge
gcloud compute security-policies rules create 1000 \
--security-policy=edge --action=deny-403 \
--src-ip-ranges=203.0.113.0/24
gcloud compute backend-services update svc --global \
--security-policy=edge
gcloud api-gateway api-configs create cfg --api=bookshelf \
--openapi-spec=openapi.yaml
gcloud api-gateway gateways create gw --api=bookshelf \
--api-config=cfg --location=$REGION
```

API Gateway: serverless OpenAPI front (x-google-backend); **Apigee:** full API management; **Endpoints:** ESPv2 sidecar.

VPN & USER ACCESS

```
gcloud compute vpn-gateways create gw --network=N --region=$REGION
gcloud compute routers create vr --network=N --region=$REGION --asn=65001
gcloud compute external-vpn-gateways create peer --interfaces 0=IP
gcloud compute vpn-tunnels create t0 --region=$REGION \
--vpn-gateway=gw --interface=0 --peer-external-gateway=peer \
--peer-external-gateway-interface=0 --ike-version=2 \
--shared-secret="$PSK" --router=vr
gcloud compute routers add-interface vr ... # then add-bgp-peer
gcloud compute start-iap-tunnel vm 22 --local-host-port=localhost:2222 --zone Z
```

HA VPN 99.99% needs 2 interfaces, 2+ tunnels, BGP. Interconnect for private capacity. No managed point-to-site: IAP TCP forwarding.

EMAIL

No first-party service; **outbound port 25 is blocked**. Use **587** (STARTTLS) with SendGrid (smtp.sendgrid.net, user apikey), Mailgun, Mailjet, the Workspace relay or the Gmail API; keys in Secret Manager. Publish **SPF, DKIM, DMARC** (_dmarc TXT).

MONITORING

```
gcloud logging read 'resource.type="cloud_run_revision" AND severity>ERROR' --freshness=1h
gcloud alpha monitoring policies create --policy-from-file=a.json
```

Logs Explorer / Log Analytics (SQL), log-based metrics, uptime checks, Trace (OpenTelemetry), **Ops Agent** on VMs. Admin Activity audit logs always on.

COST

```
gcloud billing budgets create --billing-account=ID \
--display-name=monthly --budget-amount=100USD \
--threshold-rule=percent=0.5
```

Free trial + Free Tier; committed/sustained-use discounts; Spot. Org Policy secure-by-default since 2024-05-03. Hidden drivers: Cloud NAT, HA VPN tunnels, external IPv4, LB forwarding rules, idle disks, egress, log ingestion.

RETIREMENTS & RENAMES

- Container Registry** (gcr.io): shut down 2025-03-18 → Artifact Registry.
- Cloud Functions** renamed **Cloud Run functions** 2024-08-22.
- konlet** / `create-with-container`: deprecated 2025-07-21, shutdown 2026-07-31.
- Classic VPN** BGP routing: deprecated 2025-08-01 → HA VPN.
- Deployment Manager**: end of support 2026, shutdown 2027-06-30 → Terraform.
- Memorystore for Memcached**: no new projects after 2027-02-01, shutdown 2029-01-31.
- Pub/Sub Lite**: turn-down 2027-01-31.
- Anywhere Cache** renamed **Rapid Cache** 2026-03-24.
- SendGrid** free plan retired 2025-05-27.
- Cloud Source Repositories**: closed to new customers 2024-06-17. **Cloud Debugger**: shut down 2023-05-31.